

A Classification Theorem for Admissible Standard-Model Structures

Uniqueness Under Standing-Preserving and Transport Constraints

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Abstract

This paper proves a classification theorem for admissible Standard-Model-level bookkeeping structures, assuming the previously established kernel constraints of admissibility, standing, identity-preserving redescription, and transport completeness. The target is not a new dynamical model, a derivation of numerical parameter values, or a prohibition on broader UV or BSM theories. The target is narrower: to classify the admissible bookkeeping structures capable of supporting stable, reusable Standard-Model observables.

The main result is that, once the accepted kernel constraints are applied jointly, the admissible same-regime solution space contains a single occupant up to admissible equivalence. Apparent alternatives either collapse under the standing quotient, fail transport completeness, introduce non-admissible parameter freedom, change load or scope, or reduce to skin-level redescription. The Standard Model bookkeeping structure is therefore not selected by empirical adequacy, convention, or representational preference; it is the unique minimal standing-preserving occupant of the admissible bookkeeping space.

The paper also distinguishes this result from fixed-load closure theorems: prior work establishes the admissibility kernel and same-load interior closure, whereas the present paper proves the global classification of admissible occupants in the Standard-Model regime. Remaining numerical inputs are certified as boundary data rather than as competing structures.

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1 Scope, method, and claim typing

This paper performs a forced downstream derivation under hostile-audit conditions. It does not explore alternatives by preference, appeal to empirical adequacy as a selection principle, or assert ontology. The object of derivation is a *bookkeeping occupant*: a standing-preserving structure capable of carrying identity-relevant particle-physics bookkeeping across admissible redescription, admissible regime transport, and reusable statement composition.

The central claim is conditional and exact:

Given the locked AASC forcing results listed in Section 4, every admissible standing-preserving particle-physics bookkeeping occupant is identity-preservingly equivalent to the Standard Model bookkeeping structure, with all remaining numerical freedom certified as irreducible envelope input.

Constraint role. The constraints exclude only structures that fail identity preservation, transport stability, or reuse consistency; they do not select among admissible competitors but eliminate non-admissible constructions.

Physical scope. Any physical theory capable of producing stable, reusable observational structure must instantiate an admissible bookkeeping system, and therefore lies within the classified structure.

Calibration. The strength of the result reflects the elimination of all admissibility-violating alternatives, not an expansion of assumptions.

This claim is narrower than saying that the Standard Model is the only mathematically possible quantum field theory. It is also stronger than fixed-load closure. A UV completion, BSM sector, landscape measure, boundary-condition program, nonrenormalizable extension, or alternative high-energy theory may be a valuable larger-domain construction. It is not a same-occupant counterexample unless it preserves the standing quotient, witness family, admissible transport structure, scope boundary, and load-class fixed by the locked AASC results. Any physical theory capable of reproducing Standard-Model-level observables must admit an admissible bookkeeping structure for those observables, and therefore factors through the unique structure classified here at the implementation layer.

Remark 1.1 (Hostile-audit posture). All uses of “unique,” “minimal,” “forced,” and “exhausted” are relative to the admissible standing-preserving bookkeeping space generated by the locked dependencies. The word “unique” does not mean unique across unconstrained mathematical imagination. It means unique after the relevant admissibility obligations are imposed and all inadmissible escape operators are fail-closed.

Remark 1.2 (No numerical-value overclaim). The theorem does not fix the numerical values of the remaining gauge couplings, Yukawa eigenvalues, mixing coordinates, CP-type residues, or scale anchors. It classifies their role. Numerical values not forced by standing preservation remain boundary or envelope inputs. Attempting to internalize them without a new witness, anchor, or admissible transport is a scope violation, not a completion of the same occupant.

Remark 1.3 (Physics-facing interpretation). The local physics material invoked here is standard bookkeeping material: gauge/redescription freedom, quotient-invariant parameters, charge-label rigidity, generation non-degeneracy, phase classification, and cross-regime matching. The claimed novelty is not a new Lagrangian, new dynamics, or a replacement for empirical model building. The claimed novelty is the constraint-theoretic classification: once standing, transport, non-degeneracy, scope compatibility, and repair exhaustion are imposed, the admissible bookkeeping object class has a single structural occupant.

2 Relation to *Structura Ex Necessitate Standardis Modelis*

Structura Ex Necessitate Standardis Modelis is the primary closure spine for the hardened capstone, but it is not the same theorem. It proves a fixed-load result: assuming the declared Standard-Model renormalizable chiral gauge load, fixed domain, fixed comparison class, and explicit admissibility envelope, the Standard-Model bookkeeping interior is closed under same-domain, certificate-bearing transport. It explicitly types itself as relative closure rather than a derivation of the Standard Model from first principles.

The present paper has a different target. It uses the locked upstream AASC results to identify the admissible occupant before the fixed-load interior is audited.

Result	Theorem role
<i>Structura Ex Necessitate Standardis Modelis</i>	Given the Standard-Model load, fixed scope, fixed-domain comparison class, and admissibility envelope, no further same-scope standing-bearing completion remains inside the interior.
This capstone	Given the locked AASC forcing results, only the Standard-Model bookkeeping occupant remains in the admissible standing-preserving particle-physics bookkeeping space.

Thus *Structura Ex Necessitate Standardis Modelis* supplies the interior-closure certificate needed at the final step of the uniqueness proof. The capstone supplies the occupant-level exhaustion that *Structura Ex Necessitate Standardis Modelis* deliberately does not attempt.

Definition 2.1 (Bookkeeping occupant versus bookkeeping interior). A *bookkeeping occupant* is a candidate standing-preserving structure before a particular fixed-load interior has been closed. A

bookkeeping interior is the parameter, witness, quotient, role, and transport structure internal to an occupant once its load-class has been fixed.

Proposition 2.2 (Non-redundancy of the capstone). *The present capstone is non-redundant with Structura Ex Necessitate Standardis Modelis. Structura Ex Necessitate Standardis Modelis proves closure of an interior under a declared load. The present paper proves uniqueness of the occupant relative to locked AASC forcing results. Interior closure does not imply occupant uniqueness; occupant uniqueness requires exhausting rival load-classes, charge lattices, generation patterns, phase structures, transport failures, and illicit value-fixation rules. Conversely, occupant uniqueness needs an interior-closure dependency to prevent a same-load rival after the occupant has been identified.*

Proof. If one assumes an occupant and proves no same-domain completion inside it, one has not shown that no different admissible occupant exists. If one proves that only one occupant survives the locked admissibility constraints but does not close its interior, one has not excluded same-occupant completion. These are distinct obligations. *Structura Ex Necessitate Standardis Modelis* discharges the interior obligation after the Standard-Model load-class is fixed; this paper discharges the occupant obligation by using the locked upstream forcing results. Therefore neither theorem entails the other without the missing bridge, and the capstone is not a rewrite of *Structura Ex Necessitate Standardis Modelis*. $\square$

3 Primitive interface and normal forms

This section states the theorem object precisely. It does not introduce new AASC primitives; it packages the locked dependency structure into an explicit object class so that the uniqueness claim has a determinate domain.

Definition 3.1 (Pre-admissible bookkeeping structure). A pre-admissible particle-physics bookkeeping structure is a tuple

$$\mathfrak{B} = (P, \mathcal{S}, \text{Out}, \mathcal{W}, \sim, \mathcal{T}, \mathcal{H}, \rho, \mathcal{E}),$$

where:

- (i) P is a parameter or label space for candidate bookkeeping states;
- (ii) $\mathcal{S}$ is the scope family: regimes, admitted tests, admissible approximations, and error envelopes;
- (iii) Out assigns admitted witness outputs to parameter points in skins/scopes;
- (iv) $\mathcal{W}$ is the family of invariant witnesses claimed to carry standing;
- (v) $\sim$ is the standing equivalence relation induced by equality of all admitted witness outputs within the declared error discipline;

- (vi) $\mathcal{T}$ is the class of licensed transport and redescription maps among skins and regimes;
- (vii) $\mathcal{H}$ is the continuation/history structure under which committed construction records must remain reusable;
- (viii) ρ is the role typing of components as anchor, tensor witness, or skin;
- (ix) $\mathcal{E}$ is the envelope-boundary ledger: data allowed to enter only as boundary or cross-envelope input rather than same-interior derivation.

A bookkeeping structure is not a Lagrangian alone, not a list of fields alone, and not a coordinate chart alone. It is the full witness, quotient, transport, continuation, role-typed, and envelope-typed object on which standing is asserted.

Definition 3.2 (Admissible bookkeeping structure). A pre-admissible structure $\mathfrak{B}$ is admissible when it satisfies the locked AASC obligations relevant to this target: stable identity fixation on $P/\sim$; output agreement under admitted tests; transport completeness for every licensed map in $\mathcal{T}$; preservation under admissible continuations in $\mathcal{H}$; non-degeneracy of standing witnesses; no post-hoc repair of committed records; role exhaustion under ρ into anchor, tensor witness, or skin; and boundary-only treatment of non-internally-fixed envelope inputs in $\mathcal{E}$.

Definition 3.3 (Admissible bookkeeping space). Let $\mathcal{B}_{\text{AASC}}$ denote the class of admissible particle-physics bookkeeping structures satisfying the locked constraints relevant to this target. An element of $\mathcal{B}_{\text{AASC}}$ is also called an admissible bookkeeping occupant.

Definition 3.4 (Standing-preserving occupant). An occupant $\mathcal{O} \in \mathcal{B}_{\text{AASC}}$ is standing-preserving when its tracked labels, parameters, witnesses, and records remain identity-fixed under all admissible redescrptions and licensed regime transports. A coordinate, phase, label, or numerical value that cannot be transported as an invariant witness is not standing-bearing content of the occupant.

Definition 3.5 (Identity-preserving equivalence). Two admissible bookkeeping structures $\mathcal{O}_1$ and $\mathcal{O}_2$ are identity-preservingly equivalent, written $\mathcal{O}_1 \simeq_{\text{Ad}} \mathcal{O}_2$, when there are admissible maps between them whose induced maps on standing quotients are inverse bijections and whose action preserves the witness family, role typing, envelope-boundary status, transport interface, and admissible continuation structure. Equivalence is therefore uniqueness up to skin-level redescription, not literal equality of notation, coordinate choice, gauge, basis, or presentation.

Definition 3.6 (Identity-preserving equivalence class). The identity-preserving equivalence class of an occupant $\mathcal{O}$ is

$$[\mathcal{O}]_{\text{Ad}} := \{\mathcal{O}' \in \mathcal{B}_{\text{AASC}} : \mathcal{O}' \simeq_{\text{Ad}} \mathcal{O}\}.$$

The uniqueness theorem is uniqueness of this class, not literal uniqueness of a presentation or parameter coordinate chart.

Definition 3.7 (Minimal occupant). An occupant is minimal when deletion of any standing-bearing witness, discrete label, compatibility relation, quotient component, or transport interface destroys standing preservation, non-degeneracy, witness completeness, or scope compatibility; and when any additional item either collapses as skin, changes the scope/load, or invokes a standing-failure mode.

Definition 3.8 (Boundary-only envelope input). An input is boundary-only when it is required to instantiate a witness value or cross-envelope match but is not internally fixed by same-occupant admissibility. Boundary-only input is not a rival structure and not a failure of uniqueness. It is the certified residue after structural forcing is complete.

Definition 3.9 (Exhaustion). The admissible bookkeeping space is exhausted by an occupant $\mathcal{O}_*$ when every $\mathcal{O} \in \mathcal{B}_{\text{ASC}}$ satisfies one of the following: (i) $\mathcal{O} \simeq_{\text{Ad}} \mathcal{O}_*$; (ii) $\mathcal{O}$ changes the domain, scope, load, witness family, quotient, continuation structure, or envelope boundary; or (iii) $\mathcal{O}$ collapses by a standing-failure mode.

Remark 3.10 (Interpretation statement for physics readers). The theorem is a constraint-theoretic classification result. It does not claim that local Standard-Model calculations are novel, and it does not replace dynamical model building. Its asserted contribution is the forced sorting of carrier, witness, quotient, transport, continuation, and boundary roles, followed by exhaustion of same-admissibility rivals.

4 Locked inputs and dependency ledger

The derivation proceeds strictly downstream of locked results. No dependency is weakened, extended, or reinterpreted. The identifiers below name existing locked results as used by this capstone. Each dependency eliminates a specific escape route.

Dependency	Capstone role	Escape route eliminated
SM-BK-1	Gauge structure as standing-preserving redescription bookkeeping.	Non-gauge or privileged-representation bookkeeping treated as identity-bearing without transport-complete witness content.
SM-PARAM-A	Parameter admissibility and redundancy elimination.	Free parameter surplus, generate-then-filter tuning, and post-hoc value selection.
SM-PARAM-B	Transport-closure locus for continuous parameters.	Parameters that function only locally in one skin or fail under cross-regime transport.
SM-PARAM-C	Generation multiplicity minimality and non-degeneracy.	Fewer-generation collapse and extra-generation degeneracy without standing gain.

Dependency	Capstone role	Escape route eliminated
SM-PARAM-D	Charge lattice rigidity and uniqueness.	Exotic charge assignments, continuous charge deformation, and unanchored hypercharge lattices.
SM-PARAM-E	Mixing angles as redescription coordinates.	Treating mixing-basis parameters as independent load beyond quotient-invariant flavor content.
SM-PARAM-F	CP-like phases as admissible holonomy classes.	Local CP tuning knobs and extra phase variables without holonomy standing.
SM-PARAM-G	Scale parameters and cross-envelope matching.	Internal numerical fixation of scales without boundary anchor or new tensor witness.
<i>Structura Ex Necessitate Standardis Modelis</i>	Fixed-load interior closure under declared Standard-Model load and certificate-maximal same-domain transport.	Same-load extensions, illicit same-scope completions, hidden selectors, latent labels, and post-hoc repairs inside the closed interior.

Remark 4.1 (Why the ledger matters). The baseline version of this paper stated the locked inputs but did not expose their audit function. The hardened version makes the dependency discipline explicit: every dependency removes a distinct class of rival. This prevents the theorem from appearing as an assertion that the Standard Model was simply assumed and restated.

5 Envelope necessity and non-circular input composition

The two highest-value hostile attacks are: first, that the admissibility envelope excludes competitors by stipulation; second, that the locked inputs already contain the conclusion. This section fixes both points at theorem level.

Lemma 5.1 (Envelope necessity). *A candidate construction that cannot preserve standing under identity-preserving redescription, cannot carry transport-complete witnesses across the declared regime interfaces, cannot support non-degenerate reuse of construction records, or requires post-hoc repair is not an alternative occupant of $\mathcal{B}_{\text{AASC}}$. It is a collapse of bookkeeping comparability, not a rival member of the same object class.*

Proof. A bookkeeping structure is required to compare the same standing-bearing object across skins, regimes, and admissible continuations. If identity is not preserved under lawful redescription, the object tracked in one presentation is not well defined in another. If transport is not complete, a witness distinction can be lost or created by the interface rather than carried by the object. If non-degeneracy fails, labels proliferate or collapse without standing gain. If post-hoc repair is

allowed, a committed construction record can be reindexed after the fact, destroying stable reuse. In each case the failure prevents the tuple $\mathfrak{B}$ from having a well-defined standing quotient and licensed transport structure. Therefore admissibility is not an optional filter over already comparable rivals; it is the minimum condition for a candidate to enter the comparison space at all. $\square$

Let $\mathfrak{U}$ be the pre-admissible bookkeeping universe before locked constraints are imposed. Let C_0 denote SM-BK-1, $C_1, \dots, C_7$ denote SM-PARAM-A through SM-PARAM-G, and C_S denote the fixed-load interior closure supplied by *Structura Ex Necessitate Standardis Modelis*. Write

$$\text{Fix}(C_i) = \{\mathfrak{B} \in \mathfrak{U} : \mathfrak{B} \text{ satisfies } C_i\}.$$

Definition 5.2 (Non-circular forcing family). A family of constraints $\{C_i\}$ is non-circular for a target occupant $\mathcal{O}_*$ when:

- (i) no C_i is itself the assertion $\mathcal{B}_{\text{AASC}} / \simeq_{\text{Ad}} = \{[\mathcal{O}_*]\}$;
- (ii) each C_i removes a typed rival class while leaving unresolved degrees that require other constraints;
- (iii) the joint intersection $\bigcap_i \text{Fix}(C_i)$ contains exactly the equivalence class $[\mathcal{O}_*]_{\text{Ad}}$.

Input	What it forces	What it does not by itself force
SM-BK-1	A standing-preserving redescription/gauge carrier.	It does not fix generation count, numerical values, charge lattice, mixing quotient, or CP holonomy.
SM-PARAM-A	Redundancy elimination and parameter normal form.	It does not select the carrier, charge skeleton, multiplicity, or boundary values.
SM-PARAM-B	Transport-closed locus for continuous standing.	It does not determine which discrete labels are allowed or close the fixed-load interior.
SM-PARAM-C	Minimal non-degenerate generation multiplicity.	It does not fix the charge lattice, phase classes, or gauge/redescription carrier.
SM-PARAM-D	Rigid charge-lattice class.	It does not fix generation multiplicity, mixing quotient, holonomy class, or scale boundary status.
SM-PARAM-E	Mixing as quotient/redescription coordinate structure.	It does not fix charge labels, generation count, or numerical mixing values.
SM-PARAM-F	CP-like phase structure as admissible holonomy.	It does not fix gauge carrier, charge lattice, or scale inputs.
SM-PARAM-G	Boundary-only status of scale/matching values.	It does not derive the discrete skeleton or close the occupant interior.

Input	What it forces	What it does not by itself force
<i>Structura Ex Necessitate Standardis Modelis</i>	Closure of the fixed-load Standard-Model interior.	It does not by itself identify the occupant before the load-class is forced.

Theorem 5.3 (Non-circularity of the forcing inputs). *The capstone’s uniqueness conclusion is not a premise of any single dependency. The conclusion is the joint-intersection result*

$$\bigcap_{i=0}^7 \text{Fix}(C_i) \cap \text{Fix}(C_S) = [\text{SM}_{\text{book}}]_{\text{Ad}},$$

with equality understood up to identity-preserving equivalence.

Proof. The separation table shows that each locked input removes a distinct typed failure class while leaving residual alternatives not addressed by that input alone. Thus no input is equivalent to the final singleton conclusion. The main proof below uses the inputs compositionally: carrier first, parameter normal form second, discrete skeleton third, quotient and holonomy classification fourth, boundary-only scale certification fifth, and fixed-load closure last. A rival survives a stage only if it has not yet violated the corresponding dependency. After all stages are imposed, every possible difference is either skin, domain motion, witness motion, quotient/continuation motion, or standing failure. Therefore the singleton result is the intersection of independent eliminations rather than a restatement of a hidden Standard-Model premise. $\square$

Corollary 5.4 (Referee burden for circularity objections). *A circularity objection must identify which dependency already asserts the final singleton theorem, or else show that one dependency secretly entails all the remaining ones. A generic statement that the inputs are “SM-shaped” is insufficient, because the capstone is explicitly downstream of named class eliminators and its proof depends on their joint intersection.*

Theorem 5.5 (No fixed-load backflow). *In the capstone proof, Structura Ex Necessitate Standardis Modelis is logically downstream of the occupant-forcing stage. Removing Structura Ex Necessitate Standardis Modelis leaves intact the derivation of the admissible load-class from SM-BK-1 through SM-PARAM-G, but reopens only same-load interior-completion possibilities. Therefore Structura Ex Necessitate Standardis Modelis cannot be the source of occupant uniqueness.*

Proof. The load-class forcing stage uses SM-BK-1 through SM-PARAM-G to force, respectively, the standing-preserving carrier, parameter normal form, transport-closed continuous locus, non-degenerate generation skeleton, rigid charge lattice, quotient-mixing status, holonomy-phase status, and boundary-only scale status. None of these steps invokes the fixed-load interior-closure result supplied by *Structura Ex Necessitate Standardis Modelis*. *Structura Ex Necessitate Standardis Modelis* enters only after those roles have already determined the Standard-Model bookkeeping

load-class up to admissible equivalence. If *Structura Ex Necessitate Standardis Modelis* is removed, the proof no longer closes possible same-load completions inside that already-forced occupant, but the prior carrier, charge, multiplicity, quotient, holonomy, and boundary-status forcing steps remain unchanged. Hence *Structura Ex Necessitate Standardis Modelis* supplies the final interior-closure certificate, not the occupant-forcing premise. $\square$

Remark 5.6 (Family-level circularity burden). A stronger circularity objection must therefore do more than say that *Structura Ex Necessitate Standardis Modelis* is used in the capstone. It must show that one of SM-BK-1 through SM-PARAM-G is itself a fixed-load interior-closure fragment, or that the family jointly presupposes the Standard-Model occupant rather than independently forcing its carrier, discrete skeleton, quotient structure, holonomy structure, and boundary ledger. Absent that showing, the only role left for *Structura Ex Necessitate Standardis Modelis* is downstream closure of the already-identified interior.

6 Final closure layer: necessity, category, and classification

The preceding closure layers made the dependency ledger and object class explicit. The remaining hostile route is to assert that the comparison class is still optional, that an alternative formal system can stand outside it, or that the theorem shows only consistency rather than classification. This section closes those routes.

Definition 6.1 (Formal statement system). A formal statement system is a tuple

$$\text{Stmt} = (C, \mathcal{Y}, \text{out}, \circ, \mathcal{U}, \approx),$$

where C is a class of constructions or committed records, $\mathcal{Y}$ is an output class, $\text{out} : C \rightarrow \mathcal{Y}$ is an output map, $\circ$ is a partial composition operation on C , $\mathcal{U}$ is a set of admissible context-update or transport maps $u : C \rightarrow C$, and $\approx$ is a candidate identity relation on C .

The continuation-output relation induced by Stmt is

$$c \sim_{\text{Stmt}} c' \iff \text{out}(u(c)) = \text{out}(u(c')) \text{ for every } u \in \mathcal{U}.$$

The system has well-defined reuse when future outputs do not depend on representatives of a reused record; well-defined comparison when cross-context identity is invariant under $\mathcal{U}$; and well-defined composition when $c \approx c'$ and $d \approx d'$ imply $d \circ c \approx d' \circ c'$ whenever both composites are defined.

Theorem 6.2 (Formal impossibility of non-admissible statement systems). *Let $\text{Stmt} = (C, \mathcal{Y}, \text{out}, \circ, \mathcal{U}, \approx)$ be a formal statement system satisfying well-defined reuse, well-defined comparison, and well-defined composition in the sense of Definition 6.1. If Stmt lacks identity fixation, transport consistency, quotient consistency, or equivalence-class collapse of continuation-indistinguishable*

records, then it fails to define a consistent object of evaluation. Equivalently, every consistent statement-bearing system has a well-defined standing quotient $C/\sim_{\text{Stmt}}$, and the admissible identity relation is forced to coincide with the continuation-output relation up to harmless skin refinement.

Proof. First suppose $c \approx c'$ but $c \not\sim_{\text{Stmt}} c'$. Then for some admissible update $u \in \mathcal{U}$, $\text{out}(u(c)) \neq \text{out}(u(c'))$. The future output of a reused record depends on which representative was chosen. Reuse and comparison are therefore not well defined on $C/\approx$.

Second suppose $c \sim_{\text{Stmt}} c'$ but the system retains c and c' as different standing-bearing objects. No admissible continuation-output distinguishes them. The alleged difference is therefore not an evaluable statement difference; it is a hidden label, representative privilege, or new load not present in the output structure. A system may keep such labels as skin, but it cannot treat them as standing-bearing objects of evaluation.

Third, if an update u or a composite $d \circ c$ is not congruent with the induced quotient, then later outputs depend on representative choices. Composition is then not a function on evaluated records. Thus a system satisfying reuse, comparison, and composition is forced to use the quotient by continuation-output equivalence as its identity carrier, to make updates transport-consistent on that quotient, and to collapse continuation-indistinguishable records. These are exactly the formal admissibility carriers. Non-admissible systems may exist as calculi, simulators, notations, or proposal generators; they cannot be consistent statement-bearing systems in the stated sense. $\square$

Definition 6.3 (Minimal non-degenerate compositional-evaluation system). A minimal non-degenerate compositional-evaluation system is a formal statement system in the sense of Definition 6.1 satisfying only the following structural conditions:

- (i) *composition*: some committed records can be composed or reused as inputs to later constructions;
- (ii) *comparison*: the system can determine whether two records are the same object of evaluation across at least one admissible context update;
- (iii) *irreversible reuse*: once a record has been used in a later construction, its identity carrier cannot be silently reindexed without changing the later construction record;
- (iv) *non-degeneracy*: at least one admissible continuation distinguishes some records, and at least one pair of representational variants is required to be identified as the same evaluative object.

These clauses do not mention the Standard Model, particle physics, gauge groups, charge lattices, or any AASC-specific terminology. They specify the minimal system class in which reusable statement-bearing evaluation is non-vacuous.

Theorem 6.4 (Kernel necessity as a derived structural constraint). *For every minimal non-degenerate compositional-evaluation system, the kernel obligations of admissibility are mathematically*

forced: identity fixation, admissible transport, standing equivalence, quotient consistency, and repair exclusion. More precisely, any such system that omits one of these obligations fails to define stable objects of evaluation under composition, comparison, and irreversible reuse.

Proof. Let $\text{Stmt} = (C, \mathcal{Y}, \text{out}, \circ, \mathcal{U}, \approx)$ be a minimal non-degenerate compositional-evaluation system. By composition and reuse, later records may depend on earlier records. By comparison, the system must decide when two records carry the same object of evaluation across updates in $\mathcal{U}$. By irreversible reuse, that decision cannot be changed after the reused record has entered a later construction.

If identity fixation is absent, then the same committed record can be reidentified differently across later contexts. The output of a composite record is then representative-dependent, so composition is not well defined on evaluated records. If admissible transport is absent, a record may lose or gain standing-relevant content under an allowed context update; comparison across contexts is then undefined. If standing equivalence is absent, continuation-indistinguishable records can be treated as different objects without any evaluable difference; this is hidden label retention rather than statement-bearing distinction. If quotient consistency is absent, updates and compositions fail to descend to the identity classes on which reuse depends. If repair exclusion is absent, a failed identity lineage can be reassigned after use, contradicting irreversible reuse.

The only stable identity carrier compatible with the four minimal clauses is therefore the quotient by continuation-preserving evaluative equivalence, equipped with transports that preserve that quotient and with no post-hoc repair of committed identity lineages. These are exactly the kernel obligations. Hence the kernel is not assumed as an optional framework rule; it is the minimal structural solution to the coexistence of composition, comparison, irreversible reuse, and non-degeneracy. $\square$

Corollary 6.5 (Kernel non-optional). *A denial of the kernel is not a same-domain technical alternative. It is a denial that the system under discussion supports non-degenerate reusable statement-bearing evaluation. Such a denial may define a generator, notation, simulation, or proposal space, but it does not define a comparable admissible bookkeeping occupant.*

Theorem 6.6 (No external same-type comparison class). *There is no external same-type comparison class of particle-physics bookkeeping competitors that avoids the admissibility obligations while still supplying reusable standing-bearing statements. Any purported outside class either:*

- (i) implements the same identity, witness, quotient, transport, continuation, and non-repair obligations, and hence lies inside $\mathcal{B}_{\text{AASC}}$ after retyping;*
- (ii) drops at least one of those obligations, and hence cannot define same-type reusable standing-bearing statements; or*
- (iii) introduces new load, scope, witness, anchor, or continuation structure, and hence defines a larger-domain construction rather than a same-admissibility competitor.*

Proof. Let $\mathcal{C}$ be an alleged external class of competitors. If objects of $\mathcal{C}$ support reusable statements, cross-context comparison, and composition, Theorem 6.2 forces the same standing, quotient, transport, and non-repair structure. Such objects instantiate the tuple form of Section 3 and are judged by the same admissibility obligations; the class is not external. If $\mathcal{C}$ rejects one of those carriers, then its objects cannot preserve identity or witness content under comparison and are not competitors of the same type. If $\mathcal{C}$ preserves the carriers only by adding new witnesses, tests, sectors, selectors, boundary data, or histories, then it has changed the domain being compared. These alternatives exhaust the meaning of “external comparison class.” $\square$

Definition 6.7 (Morphisms of bookkeeping structures). For admissible bookkeeping structures

$$\mathfrak{B}_i = (P_i, \mathcal{S}_i, \text{Out}_i, \mathcal{W}_i, \sim_i, \mathcal{T}_i, \mathcal{H}_i, \rho_i, \mathcal{E}_i), \quad i = 1, 2,$$

a morphism $F : \mathfrak{B}_1 \rightarrow \mathfrak{B}_2$ consists of maps on parameter classes, skins, witness ledgers, transports, and histories such that:

- (i) $p \sim_1 p'$ implies $FP(p) \sim_2 FP(p')$;
- (ii) admitted witness outputs are preserved up to the declared error discipline;
- (iii) every licensed transport/history in $\mathfrak{B}_1$ is carried to a licensed transport/history in $\mathfrak{B}_2$;
- (iv) anchor, tensor-witness, skin, and boundary-only roles are preserved;
- (v) no post-hoc representative selection or hidden label is introduced by the map.

An admissible isomorphism is a morphism admitting an inverse morphism on standing quotients, witness ledgers, and transport/history structure.

Definition 6.8 (Admissible bookkeeping category). $\mathbf{Book}_{\text{AASC}}$ is the category whose objects are admissible particle-physics bookkeeping structures and whose morphisms are admissible bookkeeping morphisms. The quotient $\mathbf{Book}_{\text{AASC}} / \cong_{\text{Ad}}$ is the class of objects up to admissible isomorphism. The main uniqueness claim is the categorical statement that this quotient has one object in the target particle-physics occupant class.

Lemma 6.9 (Transport closure and quotient groupoid). *For any fixed admissible scope S , the admissible transports among skins that preserve standing, witnesses, roles, and continuation structure form a groupoid after quotienting by standing equivalence. Identities are trivial transports, composition is ordinary composition of transport maps, closure follows from preservation of the same admissibility data, associativity follows from associativity of function composition, and inverses exist on standing quotients whenever two skins represent the same occupant. Consequently, uniqueness is asserted in the quotient groupoid of admissible skins, not in the raw presentation category.*

Proof. The identity transport on any skin preserves every standing class, witness output, role assignment, boundary ledger entry, and continuation, so identities exist. If $T_{12} : S_1 \rightarrow S_2$ and $T_{23} : S_2 \rightarrow S_3$ are admissible, then both preserve standing equivalence and all licensed witness/transport/history data. Their composite $T_{23} \circ T_{12}$ therefore also preserves those data, so admissible transports are closed under composition. Associativity is inherited from ordinary composition of maps. If two skins are asserted to represent the same occupant, admissibility requires transport completeness in both directions; after quotienting by $\sim_S$, these transports induce inverse bijections on standing classes. Thus the admissible identity-preserving transports form a groupoid on the quotient. The theorem's uniqueness claim is uniqueness of the resulting groupoid object, not uniqueness of a chosen skin. $\square$

Theorem 6.10 (Witness invariance of the standing quotient). *Let $\mathcal{W}_1$ and $\mathcal{W}_2$ be two admissible witness families for the same scope S and the same output discipline. If each family is transport-complete and determines all admitted outputs in S , then the standing equivalence relations induced by $\mathcal{W}_1$ and $\mathcal{W}_2$ coincide. Hence the standing quotient $P / \sim_S$ is invariant under admissible witness choice.*

Proof. Let $p, p' \in P$. If p and p' are equivalent under $\mathcal{W}_1$, then every witness in $\mathcal{W}_1$ agrees on p and p' . Because $\mathcal{W}_1$ determines all admitted outputs in S , every admitted output agrees on p and p' . Since $\mathcal{W}_2$ is admissible and transport-complete for the same scope, no witness in $\mathcal{W}_2$ can distinguish p and p' without inducing a difference in admitted output or transport-relevant standing content; otherwise it would introduce hidden unwitnessed structure for S . Thus p and p' are equivalent under $\mathcal{W}_2$. The reverse implication is identical with the roles of the witness families exchanged. Therefore the two induced equivalence relations are equal, and the quotient is witness-choice invariant. $\square$

Lemma 6.11 (Finite witness determinacy and constructive closure). *Admissible occupants in $\mathcal{B}_{\text{AASC}}$ are certificate-bearing structures: their witness ledgers, quotient rules, transport maps, continuation rules, role assignments, and boundary ledgers must be finitely specified or given by finitely presented evaluable schemas. Non-constructive oracle labels, non-measurable representative selections, unrestricted set-theoretic encodings, and non-evaluable completions are not additional admissible occupants.*

Proof. An admissible occupant must supply transport certificates, witness determinations, quotient comparisons, and continuation checks that can be reused in later construction records. An object whose alleged distinguishing content is accessible only through an arbitrary non-evaluable choice, oracle-coded real, non-finitely specified selector, or unrestricted set-theoretic enlargement cannot provide such certificates. If the extra structure affects no witness, quotient, transport, continuation, role, or boundary entry, it is skin. If it affects one of them without a certificate-bearing rule, it violates admissibility. Therefore pathological non-constructive variants are excluded by the same statement-system and transport-certificate requirements that define the comparison class. $\square$

Lemma 6.12 (Existence of the Standard-Model bookkeeping occupant). *The Standard-Model bookkeeping structure SM_{book} is an element of the solution set Sol : it satisfies the locked admissibility, standing, parameter-normal-form, transport-closure, charge-rigidity, generation-minimality, quotient-mixing, holonomy-phase, boundary-scale, and fixed-load interior-closure constraints used in this paper.*

Proof. Each listed condition is precisely one of the locked dependencies or the fixed-load closure dependency recorded in Section 4. SM-BK-1 supplies the redescription carrier; SM-PARAM-A and SM-PARAM-B supply parameter admissibility and transport closure; SM-PARAM-C and SM-PARAM-D supply generation and charge constraints; SM-PARAM-E and SM-PARAM-F supply quotient mixing and holonomy phase status; SM-PARAM-G supplies boundary-only scale treatment; and *Structura Ex Necessitate Standardis Modelis* supplies fixed-load interior closure. Therefore SM_{book} satisfies all defining constraints of Sol , so the solution set is non-empty. $\square$

Remark 6.13 (Compressed interpretation clarifier). Physical theories are implemented through bookkeeping structures: fields, parameters, witnesses, quotient relations, and regime transports are the carriers by which physical claims are compared and reused. The present result constrains that implementation layer. It does not add dynamics; it classifies the unique admissible standing-preserving bookkeeping occupant under the locked constraints.

Lemma 6.14 (Constraint independence and minimal generating set). *The forcing family*

$$\{\text{SM-BK-1}, \text{SM-PARAM-A}, \dots, \text{SM-PARAM-G}, \text{Structura Ex Necessitate Standardis Modelis}\}$$

is minimal for the capstone classification in the following precise sense: deleting any member reopens a non-empty typed rival class that is closed by no remaining member alone.

Proof. Delete SM-BK-1: carrier/redescription structure is no longer forced, so privileged-representation and non-gauge bookkeeping rivals reopen. Delete SM-PARAM-A: surplus parameter coordinates and tuning-knob descriptions reopen. Delete SM-PARAM-B: local but non-transportable parameters reopen. Delete SM-PARAM-C: fewer-generation collapse and extra-generation degeneracy cannot be excluded. Delete SM-PARAM-D: charge-lattice deformation and exotic charge assignments reopen. Delete SM-PARAM-E: basis-level flavor parameters can be counted as independent load. Delete SM-PARAM-F: extra local CP-like phase variables or phase conventions can be misclassified as tensor content. Delete SM-PARAM-G: internal numerical selectors and scale-value fixation reopen. Delete *Structura Ex Necessitate Standardis Modelis*: after the load-class is identified, same-load interior completions are not closed. Each reopened class is specifically the class removed by the deleted dependency and is not eliminated by merely retaining the others. Hence the listed family is minimal as a generating set for the classification theorem. $\square$

Theorem 6.15 (Classification theorem: singleton solution space). *Let Sol be the solution set of admissible particle-physics bookkeeping occupants satisfying the locked forcing family. Then*

$$\text{Sol} / \cong_{\text{Ad}} = \{[\text{SM}_{\text{book}}]_{\text{Ad}}\}.$$

Equivalently, the constraint system has exactly one admissible solution class: the Standard-Model bookkeeping occupant, up to admissible isomorphism and boundary-only envelope input.

Proof. Existence is supplied by Lemma 6.12. For uniqueness, let $\mathfrak{B} \in \text{Sol}$. By SM-BK-1, $\mathfrak{B}$ has the forced standing-preserving redescription carrier; otherwise it is not a solution. By SM-PARAM-A and SM-PARAM-B, its continuous coordinates reduce to transport-closed parameter normal form; surplus and non-transportable coordinates are not standing-bearing. By SM-PARAM-C, its generation multiplicity is the minimal non-degenerate class. By SM-PARAM-D, its charge lattice is the rigid admissible lattice class. By SM-PARAM-E, all mixing content is quotient/redescription content rather than additional independent load. By SM-PARAM-F, all CP-like phase content is holonomy-class content rather than local tuning. By SM-PARAM-G, scale and numerical matching values are boundary-only envelope inputs rather than same-interior selectors. The induced standing quotient is independent of admissible witness presentation by Theorem 6.10, and all admissible skin transports are compared in the quotient groupoid of Lemma 6.9. These conditions determine the Standard-Model bookkeeping load-class and witness normal form up to admissible isomorphism. *Structura Ex Necessitate Standardis Modelis* then closes the corresponding fixed-load interior: any same-load completion reduces to the same normal form or fails as scope/load motion, quotient motion, witness motion, continuation motion, skin, or standing failure. Therefore $\mathfrak{B} \cong_{\text{Ad}} \text{SM}_{\text{book}}$. Since Lemma 6.12 gives at least one solution and the preceding argument sends every solution to the same admissible isomorphism class, the quotient solution set is exactly the displayed singleton. $\square$

Theorem 6.16 (Formal minimality theorem). *Let SM_{book} be the unique solution class of Theorem 6.15. For any proper substructure $\mathfrak{B}' \prec \text{SM}_{\text{book}}$ obtained by deleting a standing-bearing carrier, witness, quotient component, compatibility relation, transport rule, history rule, role assignment, or boundary ledger entry, $\mathfrak{B}'$ violates at least one locked admissibility constraint. Hence no proper standing-bearing substructure is an admissible solution.*

Proof. A proper deletion either removes the redescription carrier, a parameter-normal-form constraint, a transport locus, the non-degenerate generation skeleton, the rigid charge lattice, a quotient mixing invariant, a holonomy phase class, the boundary-only scale ledger, or the fixed-load closure discipline. These cases are exactly blocked by SM-BK-1, SM-PARAM-A, SM-PARAM-B, SM-PARAM-C, SM-PARAM-D, SM-PARAM-E, SM-PARAM-F, SM-PARAM-G, and *Structura Ex Necessitate Standardis Modelis*, respectively. If the deleted item changes no witness, quotient, transport, history, role, or boundary status, it was skin rather than standing-bearing structure and the deletion was not a proper standing-bearing substructure. Thus every proper standing-bearing

substructure violates at least one constraint. □

Lemma 6.17 (Gauge structure as forced bookkeeping symmetry class). *Within this capstone, gauge structure is not used as an ontic symmetry postulate. It is the forced bookkeeping symmetry class for standing-preserving redescription: the admissible carrier of equivalence among skins whose witness content must survive identity-preserving transport. A proposed non-gauge carrier is admissible only if it supplies the same quotient-preserving transport structure, in which case it is an equivalent skin, or else it fails as a same-type bookkeeping carrier.*

Proof. SM-BK-1 licenses gauge/redescription structure as the carrier that preserves identity while allowing representational variation. If another formal carrier preserves the same standing quotient, witness ledger, role typing, and transport/history structure, then the morphism definition above makes it isomorphic to the same bookkeeping occupant. If it does not preserve those structures, it either privileges a representation, loses a witness across transport, or changes the comparison domain. Therefore the relevant role of gauge structure here is forced carrier structure, not an independently assumed physical ontology. □

Theorem 6.18 (BSM non-competition theorem). *Beyond-Standard-Model and ultraviolet completions are not same-admissibility counterexamples to the capstone unless they preserve the same load-class, witness family, standing quotient, transport/history structure, role typing, and boundary ledger. If they preserve all of these, they reduce to the same occupant in the target scope. If they change any of these, they are larger-domain constructions rather than competitors inside Sol.*

Proof. A BSM or UV proposal either adds fields, operators, symmetries, scales, histories, selectors, or boundary data, or it does not. If it does not, then it has no new standing-bearing content in the target object class and is an admissible skin or equivalent presentation after quotienting. If it does, then the added item is either anchor, tensor witness, continuation structure, or boundary ledger motion. That changes the object class being audited. The proposal may be physically valuable, but it is not a same-admissibility rival to the Standard-Model bookkeeping occupant. □

7 AASC full-power integration: impossibility, boundary, conservation, finality

The preceding sections close the conventional formal attack surface. This section deploys the full AASC architecture as an active engine rather than as background terminology. The target is stronger than ordinary singleton classification: rival generation, rival carrying, rival repair, pre-admissible escape, non-conservative variation, terminal-object avoidance, and alternate construction paths are each eliminated as typed collapses.

Definition 7.1 (AASC Impossibility Suite for the capstone). For the present particle-physics bookkeeping target, the AASC Impossibility Suite is the conjunction of three fail-closed prohibitions:

- (I1) **No generators.** No same-domain operation may generate a standing-bearing witness, label, parameter, phase, selector, or structural role that is not already carried by an admissible construction path and invariant witness ledger.
- (I2) **No carriers.** No same-domain operation may carry standing through an unlicensed, non-transport-complete, quotient-changing, or continuation-changing interface.
- (I3) **No repairs.** No failed construction lineage may be made admissible by post-hoc filtering, deferred selection, representative choice, retroactive reindexing, localized containment, or factorization after a standing-relevant commitment has already occurred.

A rival construction that violates any of (I1)–(I3) is not a second occupant; it is a typed standing failure.

Theorem 7.2 (Impossibility-suite exhaustion of rivals). *Let R be any purported non-equivalent rival to SM_{book} inside the target particle-physics comparison problem. If R is not admissibly equivalent to SM_{book} , then R violates at least one element of the AASC Impossibility Suite or leaves the same-domain target. Conversely, any construction satisfying no-generator, no-carrier, and no-repair obligations factors through the existing standing quotient and is not a non-equivalent rival.*

Proof. If R asserts a new standing-bearing witness, label, parameter, phase, sector, selector, or role not present in the forced ledger, it invokes a same-domain generator. That violates (I1) unless it is supplied as new load, new witness content, or enlarged scope; in that case it has left the same-domain target. If R asserts that the same standing content is carried through a different interface, but the interface is not transport-complete or changes quotient/continuation data, then standing is carried by an inadmissible carrier, violating (I2). If R first generates inadmissible candidates and then attempts to select, repair, filter, or reindex them into admissibility, it violates (I3). These are the only routes by which a non-equivalent same-domain rival can differ: it must add content, move content through an unlicensed interface, or repair a failed lineage. If none occurs, every alleged difference preserves the same quotient, witness ledger, transport structure, continuation structure, role typing, and boundary ledger; by the normal-form and classification results it factors through SM_{book} . $\square$

Definition 7.3 (AMetric Boundary for admissible bookkeeping). The AMETRIC Boundary is the interface between pre-admissible construction activity and admissible standing-bearing construction. Before the boundary is crossed, there is no licensed metric, ordering, typicality measure, comparison class, selector, or identity carrier with standing authority. Crossing the boundary requires a standing quotient, invariant witness ledger, transport-complete interface, continuation discipline, and non-repair of committed records.

Theorem 7.4 (AMetric Boundary forcing). *Every alleged alternative bookkeeping construction faces the following dichotomy. Either it does not cross the AMETRIC Boundary, in which case it is not a comparable standing-bearing occupant; or it crosses the boundary by supplying identity fixation, witness invariance, transport completeness, continuation stability, and repair exclusion, in which case it enters $\mathcal{B}_{\text{AASC}}$ and is governed by the singleton classification theorem. There is no third route by which a pre-admissible formalism can serve as a same-type rival while avoiding the admissibility obligations.*

Proof. A construction remaining before the AMETRIC Boundary may be a syntax, search process, simulation, heuristic generator, model proposal, or informal representation, but it lacks the licensed identity and comparison carriers required for same-type standing. It therefore cannot refute a uniqueness theorem over standing-bearing occupants. If it supplies those carriers, then it has crossed the boundary and instantiates the tuple structure of Section 3: parameter/label space, witness map, standing equivalence, transports, histories, role typing, and boundary ledger. Such an object belongs to $\mathcal{B}_{\text{AASC}}$ and is subject to Theorem 6.15. Any attempt to import a measure, selector, or comparison metric before boundary crossing uses a device that has no standing authority yet; any attempt to import it after boundary crossing moves anchor or tensor content. Both cases are already typed by the framework. $\square$

Definition 7.5 (Standing content and standing flux). For a fixed scope S , the standing content of a parameter point p is the pair

$$\Sigma_S(p) := ([p]_S, \mathcal{W}_S([p]_S)),$$

where $[p]_S$ is the standing equivalence class and $\mathcal{W}_S([p]_S)$ is its invariant witness ledger. A transformation has zero standing flux when it creates no new standing class, destroys no standing class, and changes no invariant witness ledger except by admissible identity-preserving isomorphism.

Theorem 7.6 (Standing conservation law). *Every admissible same-domain morphism in $\mathbf{Book}_{\text{AASC}}$ has zero standing flux. Consequently, a modification of the Standard-Model bookkeeping occupant either conserves standing exactly and reduces to an admissible redescription of the same occupant, or changes standing content and is therefore load motion, scope motion, witness motion, quotient motion, continuation motion, boundary motion, or standing failure.*

Proof. By definition, an admissible morphism preserves standing equivalence, admitted witness outputs, role typing, transport histories, and non-repair. Preservation of $\sim_S$ prevents creation or destruction of standing classes. Preservation of admitted witness outputs and witness-invariance prevents creation or destruction of standing-bearing tensor content. Preservation of transports and histories prevents an apparent difference from being generated by interface failure. Thus standing content is conserved under admissible same-domain maps. If a proposed modification has zero standing flux, it induces the same quotient and witness ledger and hence is an admissible redescription or skin. If it has nonzero standing flux, then some quotient class, witness, history,

role, or boundary ledger entry has changed, so the proposal has left same-occupant admissibility or collapsed. $\square$

Definition 7.7 (Normal-form quotient category). Let $\mathbf{Book}_{\text{AASC}}^{\text{NF}}$ be the quotient category obtained from $\mathbf{Book}_{\text{AASC}}$ by identifying skins related by admissible isomorphism and by replacing each admissible occupant with its standing witness normal form. Morphisms in $\mathbf{Book}_{\text{AASC}}^{\text{NF}}$ are the induced quotient-preserving normal-form maps.

Theorem 7.8 (Terminal object theorem). *In the target particle-physics component of $\mathbf{Book}_{\text{AASC}}^{\text{NF}}$, the Standard-Model bookkeeping normal form is terminal. For every admissible occupant B in the target component, there is a unique normal-form morphism*

$$q_B : B \longrightarrow \text{SM}_{\text{book}}$$

obtained by quotienting skin surplus, applying the locked witness normal form, and carrying the result through the fixed-load closure certificate. Thus the capstone is not merely a uniqueness statement over presentations; it is a final-object statement in the admissible normal-form category.

Proof. By Theorem 6.15, every admissible target occupant is admissibly isomorphic to SM_{book} . By witness invariance and the quotient groupoid lemma, this isomorphism is unique after passing to standing witness normal form: any two maps that preserve all admitted outputs, witnesses, transports, histories, role typings, and boundary entries agree on every standing class. Hence each object admits exactly one normal-form morphism into SM_{book} in the quotient category. That is precisely terminality in $\mathbf{Book}_{\text{AASC}}^{\text{NF}}$. $\square$

Definition 7.9 (Admissibly distinct construction trajectory). A construction trajectory is a finite sequence of construction records, redescriptions, transports, quotient reductions, and witness extractions leading to an occupant. Two trajectories are admissibly distinct only if, after crossing the AMETRIC Boundary, they differ in standing quotient, invariant witness ledger, transport/history structure, role typing, or boundary ledger. Differences only in syntax, order of harmless redescription, notation, gauge, basis, or pre-boundary proposal generation are not admissibly distinct.

Theorem 7.10 (Non-rederivability of an admissibly distinct Standard-Model occupant). *There is no admissibly distinct rederivation of SM_{book} inside the target AASC domain. Any construction trajectory that reaches a standing-bearing particle-physics occupant either factors through the same terminal normal form SM_{book} , or else fails before the AMETRIC Boundary, violates the Impossibility Suite, changes the domain, or produces a larger-domain construction rather than a same-occupant rival.*

Proof. Before the AMETRIC Boundary, a trajectory has no standing-bearing comparison authority, so its differences from another trajectory cannot constitute a second occupant. After the boundary, admissibility imposes standing conservation and the no-generator/no-carrier/no-repair obligations.

If two trajectories differ only by admissible redescrptions or transport order, the quotient groupoid and terminal-object theorem identify them in the same normal form. If they differ by adding content, carrying standing through an unlicensed interface, or repairing a failed lineage, they violate the Impossibility Suite. If they differ by new witnesses, anchors, scopes, histories, sectors, or boundary data, they define a larger-domain construction. Therefore no admissibly distinct same-domain trajectory yields a second Standard-Model occupant or an independent rederivation path with standing authority. $\square$

Corollary 7.11 (Full AASC capstone form). *The Standard-Model bookkeeping occupant is not only the unique minimal solution of the locked constraint system. It is the only boundary-crossed, standing-conserving, impossibility-suite-compliant, terminal normal-form occupant in the target particle-physics AASC domain, and it admits no admissibly distinct same-domain rederivation.*

8 No hidden fixed-load assumption

The strongest hostile objection is that the paper assumes the Standard Model and recovers the Standard Model. The hardened proof blocks that objection by distinguishing a raw load assumption from a locked load-class forcing chain.

Lemma 8.1 (No hidden fixed-load assumption). *This capstone does not assume the full Standard-Model load as an unargued primitive. It assumes only the locked upstream forcing results in Section 4. Those results jointly force the admissible bookkeeping roles that identify the Standard-Model load-class: gauge/redescription bookkeeping, parameter normal form, minimal non-degenerate generation structure, rigid charge lattice, quotient mixing, holonomy phase structure, and boundary-only scale anchoring.*

Proof. If the paper began with the declared renormalizable Standard-Model load and then proved only that no same-load completion remains, it would duplicate fixed-load posture. Instead, the load-class is reached by composition. SM-BK-1 supplies the admissible gauge/redescription carrier. SM-PARAM-A and SM-PARAM-B restrict admissible parameters to transport-stable normal form. SM-PARAM-C fixes multiplicity under non-degeneracy. SM-PARAM-D fixes charge-label rigidity. SM-PARAM-E and SM-PARAM-F exhaust mixing and phase freedom. SM-PARAM-G prevents illicit internal value fixation. *Structura Ex Necessitate Standardis Modelis* then closes the resulting interior. Therefore the Standard-Model load-class is the convergence point of the dependencies, not a premise silently inserted into the proof. $\square$

Remark 8.2 (What remains imported). The capstone remains downstream: it does not re-prove the locked results. The correct attack is therefore not “the capstone assumes the Standard Model,” but “one of the locked forcing results fails, is mistyped, or does not compose as claimed.” Section 15 states that burden explicitly.

9 Normal-form construction of the occupant

9.1 Discrete skeleton

SM-PARAM-C and SM-PARAM-D jointly force the discrete skeleton. Multiplicity is minimal under non-degeneracy; charge labels are rigid under standing-preserving transport. Thus a rival occupant cannot change generation multiplicity or charge assignments while remaining same-admissibility unless it supplies a new invariant witness inside the locked comparison space. Without such a witness, the change is either degeneracy, representational privilege, or load motion.

Proposition 9.1 (Discrete skeleton forcing). *Inside $\mathcal{B}_{\text{AASC}}$, every admissible occupant has the same standing-bearing discrete skeleton as the Standard-Model bookkeeping occupant, up to admitted sign, normalization, and redescription redundancies.*

Proof. By SM-PARAM-C, reducing the multiplicity collapses non-degenerate standing, while extending it adds degeneracy without standing gain. By SM-PARAM-D, deforming the charge lattice violates standing preservation or transport coherence. Any purported alternative discrete skeleton therefore changes a tensor witness, changes the load/scope anchor, or carries an unwitnessed distinction. Those are not same-admissibility alternatives. $\square$

9.2 Continuous parameter normal form

SM-PARAM-A and SM-PARAM-B prevent continuous parameters from being retained merely because a skin can name them. A continuous coordinate is admissible only if it survives as a standing-relevant transport-closed witness or as a redescription coordinate whose quotient effect is already classified.

Proposition 9.2 (Continuous normal-form exhaustion). *Inside $\mathcal{B}_{\text{AASC}}$, any admissible continuous degree of freedom is one of the following:*

- (i) *an invariant tensor witness required by the locked bookkeeping structure;*
- (ii) *a redescription coordinate classified by SM-PARAM-E;*
- (iii) *a holonomy class classified by SM-PARAM-F;*
- (iv) *a boundary-only scale or matching input classified by SM-PARAM-G;*
- (v) *skin.*

No additional continuous same-admissibility coordinate remains.

Proof. SM-PARAM-A eliminates redundancy and forces parameter normal form. SM-PARAM-B excludes non-transport-closed coordinates. SM-PARAM-E classifies mixing as redescription-coordinate structure rather than independent tensor load. SM-PARAM-F classifies CP-like phase structure as holonomy rather than local tuning freedom. SM-PARAM-G blocks internal scale fixation and assigns scale values to boundary-only matching. A coordinate outside (i)–(iv) changes no witness and is skin, or else introduces hidden load, privileged representation, or illicit transport. $\square$

9.3 Interior closure after occupant forcing

Once the preceding subsections identify the occupant/load-class, *Structura Ex Necessitate Standardis Modelis* becomes the fixed-load closure dependency. It proves that no additional same-domain completion of the Standard-Model bookkeeping interior remains after the witness ledger, quotient, continuation structure, and admissible transport envelope are fixed.

Lemma 9.3 (Interior closure dependency). *Given the occupant/load-class forced by the locked dependencies, any purported same-load completion reduces to the existing Standard-Model bookkeeping interior or is typed as load motion, scope motion, witness motion, quotient motion, continuation motion, skin, or standing failure.*

Proof. This is the fixed-load role of *Structura Ex Necessitate Standardis Modelis* in the dependency ledger. It is invoked only after the occupant/load-class has been reached by the upstream forcing dependencies. $\square$

10 Main theorem: occupant uniqueness

Theorem 10.1 (Standard Model bookkeeping uniqueness). *Within the admissible standing-preserving bookkeeping space $\mathcal{B}_{\text{AASC}}$ generated by the locked results in Section 4, every admissible bookkeeping occupant is identity-preservingly equivalent to the Standard Model bookkeeping structure, up to skin-level redescription and boundary-only envelope input. Equivalently,*

$$\mathcal{B}_{\text{AASC}} / \simeq_{\text{Ad}} = \{[\text{SM}_{\text{book}}]\}.$$

Proof. Let $R \in \mathcal{B}_{\text{AASC}}$ be an arbitrary candidate occupant.

First, R must instantiate standing-preserving redescription bookkeeping. By SM-BK-1, any occupant lacking the admissible gauge/redescription carrier cannot transport standing across lawful presentation changes and therefore fails admissibility. Hence R lies in the gauge/redescription bookkeeping class.

Second, R cannot retain arbitrary parameter surplus. By SM-PARAM-A, its parameters reduce to standing-relevant normal form; by SM-PARAM-B, continuous standing must lie on transport-closed loci. Any parameter outside that normal form is skin or standing failure.

Third, R cannot alter the discrete skeleton. By SM-PARAM-C, generation multiplicity is minimal under non-degeneracy; by SM-PARAM-D, the charge lattice is rigid. Therefore any discrete difference between R and the Standard-Model bookkeeping occupant either changes a tensor witness, changes load/scope, or introduces unwitnessed structure. None is a same-admissibility rival.

Fourth, R cannot add independent mixing or CP-like phase structure. By SM-PARAM-E, mixing freedom is exhausted by quotient/redescription coordinates; by SM-PARAM-F, CP-like phase freedom is exhausted by admissible holonomy classes. Additional phases, mixing labels, or local tunings either collapse to skin or require a new witness outside the locked space.

Fifth, R cannot internally fix or further constrain envelope-scale values. By SM-PARAM-G, scale and matching values are boundary-only inputs unless a larger domain supplies new anchors or witnesses. Same-interior value fixation is therefore an illicit selector, hidden load, or scope motion.

At this stage, R has the same forced discrete skeleton, same normal-form continuous structure, same quotient mixing classification, same holonomy phase classification, and same boundary-input status as the Standard-Model bookkeeping occupant. *Structura Ex Necessitate Standardis Modelis* then closes the interior: no further same-load identity-preserving completion remains. The full AASC layer of Section 7 closes the remaining meta-routes: a rival cannot generate new standing content, carry standing through an inadmissible interface, repair a failed lineage, remain outside the AMETRIC Boundary while competing, vary standing under an admissible morphism, or supply an admissibly distinct rederivation path. Hence R is admissibly equivalent to the Standard-Model bookkeeping occupant.

Since R was arbitrary, every occupant of $\mathcal{B}_{\text{AASC}}$ lies in the same equivalence class. Thus $\mathcal{B}_{\text{AASC}}/\simeq_{\text{Ad}}$ is singleton. $\square$

Lemma 10.2 (Minimality criterion). *Let $\mathcal{O} \in \mathcal{B}_{\text{AASC}}$ be an admissible occupant in witness normal form. If a component x of $\mathcal{O}$ is standing-bearing, then deletion of x is admissible only if all witness outputs, quotient identifications, licensed transports, and admissible continuations remain unchanged. In that case x was skin, not standing-bearing content. If any of those structures changes, deletion is not an admissible same-occupant reduction.*

Proof. Standing-bearing means that x participates in at least one invariant witness, quotient relation, transport rule, compatibility relation, or continuation condition. If deleting x changes none of these, then x had no standing role and was skin by role exhaustion. If deleting x changes any of them, the resulting object fails to preserve the same witness ledger or transport/history structure and therefore is not the same admissible occupant. These alternatives exhaust deletion cases. $\square$

Theorem 10.3 (Minimality of the Standard-Model bookkeeping occupant). *The Standard-Model bookkeeping occupant forced by the dependency ledger is minimal in $\mathcal{B}_{\text{AASC}}$. No standing-bearing part can be removed while preserving admissibility.*

Proof. The gauge/redescription carrier is required by SM-BK-1; deleting it destroys identity-preserving redescription. The parameter normal-form restrictions are required by SM-PARAM-A and SM-PARAM-B; deleting them reintroduces surplus or non-transport-closed coordinates. The generation skeleton is required by SM-PARAM-C; deleting it collapses non-degenerate standing. The charge lattice is required by SM-PARAM-D; deleting or deforming it destroys charge-label rigidity. Mixing quotient data are required by SM-PARAM-E when the relevant flavor witnesses are in scope; deleting them loses witness content, while retaining basis labels beyond them is skin. CP-like holonomy data are required by SM-PARAM-F when the relevant phase witnesses are in scope; deleting them loses holonomy standing, while adding local phase selectors is inadmissible. Boundary-only scale treatment is required by SM-PARAM-G; deleting it either leaves values untyped or reintroduces illicit value fixation. Finally, *Structura Ex Necessitate Standardis Modelis* closes the fixed-load interior, so no same-load reduction/extension remains except skin. By the minimality criterion, the occupant is minimal. $\square$

Corollary 10.4 (Exhaustion). *No alternative or extended same-admissibility bookkeeping regime remains. Any alleged non-equivalent rival must deny at least one locked dependency or leave the admissible comparison space.*

11 Rival exhaustion theorem

The previous theorem can be audited by rival class. The following table records the explicit collapse map. It is deliberately concrete: a referee proposing a rival must identify which row is escaped and which locked dependency is denied.

Rival class	Collapse diagnosis	Blocking dependency
Different object definition	If it lacks parameter space, witnesses, quotient, transports, histories, or role typing, it is not the theorem object; if it has them, it is judged by the same admissibility obligations.	Bookkeeping-structure definition; envelope necessity
Alternative non-gauge bookkeeping	Cannot carry standing-preserving redescription across admissible skins without privileged representation or loss of transport.	SM-BK-1
Different gauge carrier with same outputs	If quotient and transports are preserved, it is an equivalent skin; if not, it changes witness/transport structure.	SM-BK-1; identity-preserving equivalence

Rival class	Collapse diagnosis	Blocking dependency
Different representation skeleton	Changes the load-class or requires new standing witnesses; if no witness changes, it is skin.	SM-BK-1; SM-PARAM-C/D
Exotic charge lattice	Violates charge-lattice rigidity or moves load/scope.	SM-PARAM-D
Continuous charge deformation	A deformation not witnessed is skin; a witnessed deformation changes tensor content and leaves the same-admissibility class.	SM-PARAM-D; role exhaustion
Different generation count	Fewer generations collapse non-degeneracy; extra generations add degeneracy without standing gain unless new load/scope is declared.	SM-PARAM-C
Extra continuous parameter	Not in transport-closed normal form; either skin, hidden load, or a new witness outside the fixed space.	SM-PARAM-A, B
Non-transportable local parameter	Works in one skin but fails cross-regime comparison, so it cannot carry standing.	SM-PARAM-B
Extra mixing parameter	Quotient/redescription surplus beyond invariant mixing content.	SM-PARAM-E
Alternative flavor basis counted as structure	Basis choice is skin unless it changes quotient-invariant flavor witnesses, in which case it moves tensor content.	SM-PARAM-E
Extra CP-like local phase	Not an admissible holonomy class; local tuning is skin or standing failure.	SM-PARAM-F
Alternative phase convention	Rephasing convention is skin if witnesses are unchanged; otherwise it changes holonomy content.	SM-PARAM-F
Internal scale-value selector	Attempts same-interior value fixation; imports boundary data, measure, typicality, or privileged representative.	SM-PARAM-G
Numerically different couplings	Different values instantiate the same occupant with different envelope inputs; they do not define a second structure.	SM-PARAM-G; irreducible input lemma
Additional field or sector	If no new witness, redundancy for the scope; if new witness, load/scope change rather than same-admissibility rival.	Added-load dichotomy; <i>Structura Ex Necessitate Standardis Modelis</i>
Higher-dimensional operator as primitive	If it is a transport image, it is skin/EFT book-keeping; if primitive, it adds load or scope.	SM-PARAM-B/G; <i>Structura Ex Necessitate Standardis Modelis</i>
Selector, measure, or typicality repair	Canonical representative or measure is anchor, tensor, or skin; never a fourth same-scope authorizer.	Parameter normal form; <i>Structura Ex Necessitate Standardis Modelis</i>

Rival class	Collapse diagnosis	Blocking dependency
Post-hoc repair of failed lineage	Later filtering cannot retroactively create standing in the same construction record.	Repair exhaustion; <i>Structura Ex Necessitate Standardis Modelis</i>
Latent hidden label	No same-scope standing without invariant witness; with witness it changes the tensor ledger.	Standing quotient discipline
Hidden-variable encoding	If it changes no admitted witness, it is skin/hidden load; if it changes one, the domain is enlarged.	Envelope necessity; role exhaustion
Nonlocal or dual formalism	If transport-complete and quotient-preserving, it is equivalent; if not, it is not a same-occupant map.	Identity-preserving equivalence; <i>Structura Ex Necessitate Standardis Modelis</i>
History/path-integral reindexing	Cannot reassign committed records after use while preserving same-domain identity.	Continuation stability; repair exhaustion
BSM or UV completion presented as same-scope rival	Either reduces to the same interior in the low-energy scope or changes load, witness family, continuation structure, or boundary data.	SM-PARAM-G; <i>Structura Ex Necessitate Standardis Modelis</i>
Empirical-equivalent alternative model	Output agreement alone is insufficient; it must also preserve quotient, transport, histories, and role typing.	Envelope necessity; bookkeeping-structure definition

Theorem 11.1 (Rival exhaustion). *The rival classes in Section 11 exhaust the possible same-admissibility escape routes. Any alleged rival must be one of: different object definition, alternative carrier, alternative discrete skeleton, additional continuous structure, additional quotient/mixing structure, additional holonomy/phase structure, scale-value selector, extra load, hidden label, repair operator, or larger-domain completion. Each class is blocked by the dependency ledger or is typed as outside the same-admissibility space.*

Proof. A non-equivalent rival must differ from the Standard-Model bookkeeping occupant somewhere. By role exhaustion, the difference must move an anchor, a tensor witness, or skin. If it moves an anchor, it is a load/scope/envelope change. If it moves a tensor witness, it appears as a discrete-skeleton, continuous-parameter, mixing, phase, scale, or new-sector difference; these are precisely the rows in the table. If it moves only skin, it is not a non-equivalent occupant. If it asserts a fourth role, it is a selector, hidden label, repair operator, or representational privilege, again covered above. If it denies the theorem object itself, it either lacks the quotient/transport/witness machinery needed for comparison or is re-entered into the same table as a pre-admissible bookkeeping structure. $\square$

12 Witness ledger for the hardened capstone

The capstone should not leave the reader guessing which components are forced, which are quotient data, and which remain boundary input.

Structure	Standing status	Forced by	Residual freedom
Gauge/redescription bookkeeping	Carrier of admissible local bookkeeping	SM-BK-1	Gauge/basis skin only
Discrete generation skeleton	Minimal non-degenerate classification	SM-PARAM-C	None inside same scope
Charge lattice	Rigid bookkeeping labels	SM-PARAM-D	Sign/normalization redundancy only
Continuous parameter normal form	Transport-stable witnesses only	SM-PARAM-A, B	Boundary values where certified
Mixing data	Quotient/redescription invariants	SM-PARAM-E	No extra flavor tensor load
CP-like phase data	Holonomy classes	SM-PARAM-F	No independent local phase tuning
Scale and matching values	Boundary-only envelope input	SM-PARAM-G	External matching, not same-interior prediction
Full interior	Closed same-load witness, quotient, and transport structure	<i>Structura Ex Necessitate Standardis Modelis</i>	No same-load completion
Numerical witness values	Irreducible input where not forced	Boundary/envelope certification	Not predicted by this theorem

13 Certification of irreducible envelope inputs

A uniqueness theorem for the bookkeeping occupant is compatible with irreducible envelope inputs. This is essential: otherwise the capstone would overclaim and become vulnerable to the objection that it does not predict numerical constants.

Definition 13.1 (Irreducible envelope input). An irreducible envelope input is a value required to instantiate the unique occupant but not internally fixed by same-interior admissibility. It may be supplied by boundary matching, larger-domain physics, empirical calibration, or an external value-selection program. Within the capstone, it is certified as non-derivable without new anchor or tensor load.

Proposition 13.2 (No same-interior numerical value fixation). *The hardened uniqueness theorem does not internally fix the numerical values of gauge couplings, Yukawa eigenvalues, mixing angles,*

Higgs-sector parameters, scale values, or residual CP-like boundary data. Any same-interior attempt to fix such values requires a selector, measure, typicality postulate, privileged representation, compensating load, or silent boundary motion.

Proof. By SM-PARAM-G, scale and matching values are boundary-only unless a larger envelope supplies additional witnesses. By parameter normal form, an irreducible witness is retained precisely because its value is not eliminated by admissible redescription or transport. A same-interior value selector must therefore add structure. If that structure changes the domain, it is anchor motion; if it changes witness content, it is tensor motion; if it changes neither, it is skin and cannot select standing. These cases exhaust the possibilities. $\square$

Lemma 13.3 (Irreducible inputs do not multiply occupants). *Distinct numerical assignments to irreducible envelope inputs instantiate different parameter points or boundary-matched models within the same bookkeeping occupant; they do not create distinct occupants unless the assignment changes the witness family, standing quotient, role typing, transport structure, or envelope boundary.*

Proof. An occupant is the structural bookkeeping object: quotient, witnesses, transports, roles, continuations, and envelope typing. A numerical value of an irreducible input fills a coordinate or boundary slot inside that object. If changing the value leaves the object type, admitted witness family, and transport rules fixed, the structure has not changed; only an instance has changed. If changing the value requires a new selector, boundary condition, field sector, witness, or transport rule, then the domain has changed and the result is a larger-domain construction rather than a second same-admissibility occupant. Thus irreducible values are compatible with singleton occupant uniqueness. $\square$

Remark 13.4 (Why this strengthens rather than weakens uniqueness). Certifying irreducible inputs prevents illicit overreach. The paper proves uniqueness of the admissible bookkeeping occupant, not omniscient prediction of every envelope value. A larger-domain program may attempt value fixation, but then it changes the domain whose closure is being audited.

14 Relation to BSM, UV completions, and larger-domain programs

The capstone is not a prohibition on beyond-Standard-Model model building. It is a typing theorem. Let ΔL be proposed added structure.

Proposition 14.1 (Added-load dichotomy). *For a fixed admissible scope, an added sector ΔL either introduces no new invariant witness for that scope, in which case it is redundant/skin relative to that scope, or it introduces a new invariant witness, in which case the scope/load has changed. In neither case is ΔL a same-admissibility rival occupant.*

Proof. If ΔL leaves every witness output and continuation structure unchanged, it changes no standing class and factors through the same normal form. If it changes witness output, admitted tests, boundary data, or continuation structure, it has moved the domain. Same-admissibility rivalry requires non-equivalence without domain motion; neither alternative supplies that. $\square$

Thus a UV completion may be physically valuable, explanatory, or empirically motivated. It is simply not a counterexample to the capstone unless it preserves the exact same admissibility interface and nevertheless produces a new standing-bearing occupant. The main theorem says that cannot occur without denial of a locked dependency.

15 Counterexample burden and hostile-referee attack map

A genuine counterexample must do all of the following simultaneously:

- (C1) preserve the locked admissibility interface;
- (C2) preserve standing under identity-preserving redescription;
- (C3) preserve transport closure across admissible regime interfaces;
- (C4) preserve non-degeneracy and scope compatibility;
- (C5) preserve the forced charge lattice and multiplicity constraints, or refute the named locked results;
- (C6) preserve the quotient classification of mixing and holonomy classification of phases;
- (C7) avoid internal value fixation by selector, measure, typicality, or privileged representative;
- (C8) avoid load/scope/witness/quotient/continuation motion;
- (C9) exhibit an explicit invariant witness if any non-equivalent standing difference is claimed;
- (C10) show why *Structura Ex Necessitate Standardis Modelis* does not close the resulting fixed-load interior;
- (C11) show why the proposed structure is a non-isomorphic tuple $\mathfrak{B}$ rather than a different presentation or parameter instance;
- (C12) identify which constraint predicate is denied, bypassed, or shown not to compose;
- (C13) show how the proposal avoids the Impossibility Suite without generating, mis-carrying, or repairing standing;
- (C14) show how it crosses the AMETRIC Boundary while remaining outside the singleton classification;

- (C15) show how it changes neither standing content nor domain while remaining non-equivalent;
- (C16) show why terminal-object finality and non-rederivability do not force factorization through SM_{book} .

Attack	Required response
“This duplicates <i>Structura Ex Necessitate Standardis Modelis</i> .”	Denied by the occupant/interior distinction. <i>Structura Ex Necessitate Standardis Modelis</i> closes a fixed-load interior; this paper proves occupant uniqueness from locked forcing dependencies.
“You assumed the Standard Model.”	The paper assumes the locked forcing results, not the raw Standard-Model load. The load-class is reached by dependency composition.
“The locked inputs already contain the conclusion.”	Denied by the non-circularity theorem: each input controls a distinct escape class and leaves other degrees of freedom open.
“Your admissibility envelope excludes rivals by definition.”	Denied by envelope necessity: without a standing quotient and transport-complete comparison, the candidate is not a comparable bookkeeping occupant.
“The object class is undefined.”	The bookkeeping structure is the explicit tuple $(P, \mathcal{S}, \text{Out}, \mathcal{W}, \sim, \mathcal{T}, \mathcal{H}, \rho, \mathcal{E})$; uniqueness is over identity-preserving equivalence classes of that tuple.
“Non-admissible systems are possible.”	Possible as formal calculi or proposal generators, but Theorem 6.2 shows they cannot simultaneously support reuse, cross-context comparison, and compositional statement-bearing evaluation.
“Your transports are not categorically controlled.”	Lemma 6.9 proves identity, closure, associativity, and quotient inverses for admissible transports among skins.
“Uniqueness depends on witness choice.”	Theorem 6.10 proves that any transport-complete witness family determining the same admitted outputs induces the same standing quotient.
“You prove uniqueness but not existence.”	Lemma 6.12 supplies non-emptiness before Theorem 6.15 proves singleton classification.
“The gauge group is not derived here.”	Correct as a local calculation; its admissible bookkeeping role is imported through SM-BK-1 and the locked parameter skeleton. The capstone is downstream.
“This forbids BSM physics.”	No. BSM is typed as load/scope/witness extension unless it reduces to the same normal form in the fixed scope.
“The numerical parameters are not predicted.”	Correct. They are irreducible envelope inputs unless larger-domain anchors or witnesses are supplied.

Attack	Required response
“There could be another admissible skin.”	It must supply a transport-complete certificate. If it preserves the quotient it factors through the same normal form; if not, it moves the domain or fails standing.
“A hidden variable could distinguish cases.”	Not same-scope standing unless it appears as an invariant witness. If it appears, witness content changes; if it does not, it is skin or hidden load.
“A selector or typicality rule can fix values.”	The selector is anchor, tensor, or skin. As same-interior authority it is representational privilege or illicit scope transport.
“Extra generations or exotic charges are possible.”	As larger-domain model building, perhaps. As same-admissibility occupants, they violate SM-PARAM-C or SM-PARAM-D unless a locked dependency is refuted.
“Mixing and phases are physical, not skin.”	The paper does not deny physical witnesses. It classifies admissible mixing as quotient/redescription invariant and CP-like phases as holonomy classes, not extra independent tensor load.
“A rival can stay before admissibility and still compete.”	No. The AMETRIC Boundary theorem says pre-admissible constructions lack standing-bearing comparison authority; after boundary crossing they enter $\mathcal{B}_{\text{AASC}}$.
“A construction can generate new standing content.”	The Impossibility Suite blocks same-domain generators. New standing content is new witness/load/scope, not a same-occupant rival.
“A different formalism carries the same content without your transports.”	A same-content carrier must be transport-complete and quotient-preserving, hence belongs to the admissible groupoid; otherwise it violates the no-carrier clause.
“A failed construction can be repaired by later selection.”	The no-repair clause and non-rederivability theorem block post-hoc filtering, representative choice, and retroactive lineage repair.
“There may be an independent derivation of the same structure.”	As a proof in ordinary prose, perhaps; as an admissibly distinct construction trajectory, no. The trajectory either factors through the terminal normal form or changes domain/collapses.
“The result is merely unique, not final.”	The terminal-object theorem strengthens uniqueness: in the normal-form quotient category every admissible target occupant has a unique morphism into SM_{book} .
“The theorem is too strong.”	Its strength is typed: singleton occupant in $\mathcal{B}_{\text{AASC}}$, not global metaphysical impossibility over all formal theories.

16 The hardened capstone sequent

The proof can be compressed into a dependency sequent. To avoid treating a fixed-load closure theorem as occupant uniqueness, the sequent has two stages:

SM-BK-1 through SM-PARAM-G

$\implies$ unique admissible Standard-Model bookkeeping load-class,

unique load-class $\wedge$ Structura interior closure

$\implies \mathcal{B}_{\text{AASC}} / \simeq_{\text{Ad}} = \{[\text{SM}_{\text{book}}]\},$

Kernel necessity $\wedge$ Impossibility Suite $\wedge$ AMETRIC Boundary $\wedge$ standing conservation

$\implies \text{SM}_{\text{book}}$ is terminal and admits no admissibly distinct rederivation.

This is the full-power capstone claim. It is not satisfied by *Structura Ex Necessitate Standardis Modelis* alone. It is satisfied only when the upstream forcing results first identify the unique admissible occupant-class, *Structura Ex Necessitate Standardis Modelis* closes the interior of that occupant, and the AASC full-power layer rules out generators, inadmissible carriers, repairs, pre-boundary competitors, non-conservative standing variation, and independent same-domain rederivation paths.

17 Conclusion

The Standard Model bookkeeping structure is the unique minimal occupant of the locked admissible standing-preserving bookkeeping space. The proof is not a fixed-load replay of *Structura Ex Necessitate Standardis Modelis* and not a numerical prediction theorem. It is a composed exhaustion theorem and a full-power AASC closure theorem. Gauge/redescription bookkeeping supplies the admissible carrier; parameter normal form removes surplus; transport closure restricts continuous standing; non-degeneracy fixes multiplicity; charge-lattice rigidity fixes labels; quotient classification exhausts mixing; holonomy classification exhausts CP-like phase structure; boundary-only anchoring prevents illicit value fixation; fixed-load interior closure removes remaining same-domain completions; the Impossibility Suite blocks autonomous generators, independent carriers, and same-lineage repairs; the AMETRIC Boundary blocks pre-admissible competitors; standing conservation blocks same-domain variation; terminal-object finality gives the reduced categorical normal form; and non-rederivability blocks admissibly distinct construction paths to a second endpoint.

Therefore any proposed non-equivalent rival must either deny a named locked dependency, move load or scope, change the witness family, alter the standing quotient, alter the continuation/transport structure, introduce hidden unwitnessed labels, invoke a repair/selector operation, fail to cross

the AMETRIC Boundary, violate no-generator/no-carrier/no-repair, fail standing conservation, or produce only an equivalent derivation path to the terminal normal form. None is a same-admissibility Standard-Model bookkeeping rival. The admissible occupant space is singleton up to skin-level redescription and boundary-only envelope input. The final closure lock fixes the remaining audit surface: the object class is explicit, non-admissible statement systems are impossible under reuse/comparison/composition, admissible transports form the relevant quotient groupoid, the standing quotient is witness-invariant, existence is stated before singleton classification, the input composition is non-circular, minimality is componentwise, irreducible numerical inputs are certified as instances rather than rival occupants, and the full AASC architecture is active, and the kernel is stated explicitly as a mathematically derived necessity constraint rather than a free framework assumption.

18 Uniqueness of the standing equivalence relation

Theorem 18.1 (Uniqueness of the standing equivalence relation). *Fix a scope S with admitted witness-output family $\mathcal{W}_S$. Let $\equiv$ be any admissible identity relation on the relevant presentation space satisfying witness soundness, witness completeness, transport stability, and reuse consistency. Then $\equiv$ coincides uniquely with the standing equivalence relation $\sim_S$:*

$$p \equiv q \iff p \sim_S q.$$

Hence the standing quotient $P(S)/\sim_S$ is not a chosen quotient but the unique admissible identity quotient for the scope.

Proof. Witness soundness gives $p \equiv q \Rightarrow p \sim_S q$, since admissibly identical presentations cannot be separated by any admitted witness. Witness completeness gives $p \sim_S q \Rightarrow p \equiv q$, since equality of all admitted witness outputs is exactly the condition for standing equivalence in scope S . Therefore the two relations contain the same ordered pairs. Transport stability and reuse consistency rule out replacing $\sim_S$ by a weaker, stronger, or non-transport-stable relation: a weaker relation identifies witness-distinct states, a stronger relation separates witness-indistinguishable states, and a non-stable relation makes future admissible reuse representative-dependent. Thus $\sim_S$ is the unique admissible identity relation. $\square$

Corollary 18.2 (No alternative identity quotient). *No alternative same-scope identity quotient exists. Any proposed identity relation distinct from $\sim_S$ either identifies states separated by an admitted witness, separates states not separated by any admitted witness, fails transport stability, or fails reuse consistency. Consequently, all admissible same-scope identity structure factors through the standing quotient.*

A Translation to standard mathematical language

This appendix expands and supersedes the short translation note used in the positioning-lock version. It is included for reader orientation only. The correspondences below are not replacements for the terminology of the main text. The AASC terms impose additional structural requirements beyond their nearest standard mathematical analogues, especially transport completeness, witness-completeness, and reuse consistency. Thus the paper may be read in ordinary quotient-and-morphism language, but the stronger AASC conditions remain the operative constraints.

AASC term	Standard mathematical analogue	Additional constraint carried here
Standing	Equivalence-class identity; quotient object	The equivalence class must be fixed by admitted witnesses, stable under admissible transport, and reusable under later construction without representative-dependence.
Standing equivalence $\sim_S$	Equivalence relation on a presentation or parameter space	It is the unique admissible identity relation for scope S : it identifies exactly witness-indistinguishable states and no more.
Standing quotient $P(S)/\sim_S$	Quotient space, quotient set, or quotient object	It carries all and only standing-bearing content for the scope; no canonical representative is privileged.
Admissibility	Consistency plus invariance	It is fail-closed: a construction must preserve identity, witness outputs, transport, and reuse. Post-hoc repair, silent redescription, and non-transportable structure are not admissible.
Admissible transport	Structure-preserving morphism	The morphism must preserve the standing quotient and admitted witness content; when identity is asserted across presentations it must be invertible on the quotient.
Admissibility groupoid	Groupoid of quotient-preserving morphisms	Objects are admissible skins/presentations for a fixed scope; morphisms are transports inducing isomorphisms of standing quotients.
Bookkeeping skin	Presentation, coordinate system, gauge, basis, parametrization, or representational model	A skin has no standing authority unless it changes an admitted witness; otherwise it factors through the same quotient.
Bookkeeping structure	Tuple $(P, \mathcal{W}, \sim, \mathcal{T}, \circ)$ of presentations, witnesses, identity relation, transports, and composition	The tuple is admissible only if witnesses determine the quotient, transports preserve it, and composition/reuse are well-defined on it.

AASC term	Standard mathematical analogue	Additional constraint carried here
Tensor witness	Invariant coordinate or observable datum	A tensor witness changes the standing quotient or admitted outputs; it is not presentation-level surplus.
Skin	Redundant representative-level data	Skin changes notation, gauge, basis, or parametrization without changing any witness or quotient class.
Anchor	Domain-of-definition datum	Load, scope, admitted tests, error envelope, and transport interface are anchors; moving one changes the problem rather than refining the same-scope structure.
Kernel	Minimal necessity constraint set	The kernel is derived from the possibility of non-degenerate composition, reuse, and comparison; it is not an empirical postulate or optional convention.
Impossibility Suite	Obstruction/no-go results	The no-generator, no-carrier, and no-repair clauses rule out autonomous creation of standing, association-based transfer of standing, and post-failure same-lineage repair.
Standing conservation	Invariance under morphisms	Admissible transformations preserve standing content; they do not create, amplify, transfer, or recover it.
AMETRIC Boundary	Domain-of-definition boundary	Pre-admissible proposal generation does not yet supply quotient-bearing objects; to enter the evaluated domain it must instantiate the admissibility tuple.
Terminal object	Terminal object in a quotient category	The normal-form Standard-Model bookkeeping quotient receives a unique admissible morphism from each same-scope admissible presentation.
Non-rederivability	No admissibly distinct construction path to a non-equivalent endpoint	Alternative derivations either factor through the same quotient normal form, change domain, or collapse by inadmissibility.
Collapse	Quotient reduction or obstruction-triggered rejection	Collapse is not rhetorical; it means either reduction to the same quotient class or failure of an admissibility condition.
Envelope	Domain/category of admissible constructions	The envelope is maximal relative to identity fixation, witness completeness, transport stability, and reuse consistency; outside-envelope moves are typed as scope/load changes unless recertified.

Reading rule. Under this translation, the main theorem can be read as a quotient-and-invariance classification result. The Standard-Model bookkeeping occupant is the unique admissible quotient object satisfying the locked witness, transport, non-degeneracy, charge, phase, scale, and closure constraints, up to structure-preserving isomorphism and boundary-only envelope inputs.

Non-reduction warning. The translation is intentionally one-way. An ordinary quotient, morphism, or invariant need not satisfy the full AASC requirements. In particular, conventional quotient language alone does not require transport completeness, reuse consistency, or fail-closed rejection of post-hoc repair. The paper uses the orthodox analogues to make the mathematical form legible; it uses the stronger AASC conditions to do the closure work.

Non-optional strengthening. The strengthened constraints are not optional enrichments but are forced by the necessity of identity stability, transport consistency, and reuse under composition; removing them collapses the admissible system.

Physical-scope clarification. The result constrains the admissible form of any theory capable of producing Standard-Model-level witness structure. It does not assert that larger-domain theories, ultraviolet completions, or beyond-Standard-Model loads cannot exist; it asserts that such theories must either factor through the unique Standard-Model bookkeeping quotient at the SM scope or else move load, scope, witnesses, or transport structure.

Calibration. The strength of the result reflects the elimination of all admissibility-violating alternatives, not an expansion of assumptions. The apparent strength is a consequence of the constraint structure: once the admissible object class is fixed, the rival same-scope solution space is exhausted.

B Canon integration checklist

For corpus integration, this paper should be treated as a Standard-Model uniqueness capstone, not as a replacement for *Structura Ex Necessitate Standardis Modelis*. The proper dependency direction is:

locked forcing results $\longrightarrow$ this uniqueness capstone,
this uniqueness capstone $\longrightarrow$ use of Structura as interior-closure dependency.

If the canon requires filename-level cross-reference, this paper should cite *Structura Ex Necessitate Standardis Modelis* explicitly as the fixed-load interior-closure theorem and should cite charge-lattice rigidity, generation-count bounds, gauge bookkeeping, parameter admissibility, mixing, holonomy,

and scale-boundary papers as locked upstream dependencies. In the AASC architecture layer, it should also be indexed under Impossibility Suite deployment, standing conservation, AMETRIC Boundary forcing, terminal-object closure, and non-rederivability.

C Minimal referee checklist

A referee report should answer these questions before classifying the theorem as redundant or false:

1. Is the objection aimed at occupant-level uniqueness or fixed-load interior closure?
2. Which locked dependency is denied or mistyped?
3. Does the proposed rival preserve the same load, scope, witness family, quotient, transport interface, and continuation structure?
4. If it is non-equivalent, what invariant witness distinguishes it?
5. If it fixes a numerical value, what anchor or tensor witness supplies that selector?
6. If it invokes BSM or UV structure, why is that not load/scope motion?
7. If it invokes a hidden label, how does the label acquire same-scope standing?
8. If it invokes a new skin, why does it not factor through the same normal form?
9. If it invokes a new generator, carrier, or repair, which branch of the Impossibility Suite is not violated?
10. If it invokes a pre-admissible alternative, how does it cross the AMETRIC Boundary without instantiating the admissibility tuple?
11. If it invokes an alternate construction trajectory, why does it not collapse to the terminal normal form?

A report that does not answer these questions has not supplied a same-admissibility counterexample.

References

- [1] Amos Jay Maley, *Structura Ex Necessitate Standardis Modelis: A Relative Closure Theorem for the Standard-Model Bookkeeping Interior under Fixed Load, Fixed Domain, and Certificate-Maximal Same-Domain Transport*, AASC corpus manuscript, 2026.

- [2] Amos Jay Maley, *Gauge Structure as Standing-Preserving Redescription Bookkeeping*, locked AASC dependency cited as SM-BK-1.
- [3] Amos Jay Maley, *Parameter Admissibility and Redundancy Elimination*, locked AASC dependency cited as SM-PARAM-A.
- [4] Amos Jay Maley, *Transport-Closure Locus for Continuous Parameters*, locked AASC dependency cited as SM-PARAM-B.
- [5] Amos Jay Maley, *Generation Multiplicity Minimality and Non-Degeneracy*, locked AASC dependency cited as SM-PARAM-C.
- [6] Amos Jay Maley, *Charge Lattice Rigidity and Uniqueness*, locked AASC dependency cited as SM-PARAM-D.
- [7] Amos Jay Maley, *Mixing Angles as Redescription Coordinates*, locked AASC dependency cited as SM-PARAM-E.
- [8] Amos Jay Maley, *CP-like Phases as Admissible Holonomy Classes*, locked AASC dependency cited as SM-PARAM-F.
- [9] Amos Jay Maley, *Scale Parameters and Cross-Envelope Matching*, locked AASC dependency cited as SM-PARAM-G.
- [10] Amos Jay Maley, *AASC Impossibility Suite: No Generators, No Carriers, and No Repairs*, AASC architecture dependency invoked in the full-power capstone layer.
- [11] Amos Jay Maley, *Standing Conservation under Admissible Construction*, AASC architecture dependency invoked for conservation-form uniqueness.
- [12] Amos Jay Maley, *The AMetric Boundary as a Regime-Invariant Constraint on Admissibility-Bearing Construction*, AASC architecture dependency invoked for boundary forcing.
- [13] Amos Jay Maley, *Non-Rederivability and Terminal Closure of Admissible Construction Paths*, AASC architecture dependency invoked for path-collapse and terminality.

Uniqueness of the Standing Equivalence Relation

Theorem C.1 (Uniqueness of the Standing Equivalence Relation). *Fix a scope S with admitted witness-output family $\mathcal{W}_S$. Let $\equiv$ be any admissible identity relation satisfying witness soundness, witness completeness, transport stability, and reuse consistency. Then $\equiv$ coincides uniquely with the standing equivalence relation $\sim_S$.*

Proof. Witness soundness implies $\equiv \Rightarrow \sim_S$. Witness completeness implies $\sim_S \Rightarrow \equiv$. Hence they coincide. Transport stability and reuse consistency ensure uniqueness. $\square$

Corollary C.2 (No Alternative Identity Quotient). *No alternative same-scope identity quotient exists.*